

# PROJECT INCREMENT & PROGRESS REPORT

Doton City Pop Discography & Community Vault · Course Code: 6APSI

|                                             |                                                 |
|---------------------------------------------|-------------------------------------------------|
| Project Phase: Midterm / Pre-Public Audit   | Date: September 23, 2026                        |
| Repository Status: Public Lockdown Prepared | Early Start Status: Initiated Ahead of Schedule |

## ★ EARLY INITIATION ADVANTAGE & STRATEGIC HIGHLIGHT

**Proactive Timeline:** Work on this project was **started early in the preliminary cycle**, well ahead of standard submission schedules. Initiating early provided significant strategic technical advantages: it allowed ample time for thorough architectural planning, hand-crafting a modular design system (DOTON\_DESIGN\_SYSTEM.md), integrating a dual dynamic/fallback Supabase data layer, and completing all pre-public security lockdown requirements (SECURITY-CHECKLIST.md) before public repository release.

## 1. Executive Summary

The **Doton City Pop Vault** platform is a full-stack web application designed to preserve, showcase, and recommend Japanese 1970s and 1980s City Pop vinyl albums. By combining a modern React frontend with a PostgreSQL/Supabase cloud database, the application enables users to filter albums by mood/vibe, play 30-second audio previews continuously across viewports, and submit new community album recommendations in real time.

Due to early commencement of development, all core functional requirements and security milestones have been completed cleanly ahead of deadline. The codebase has undergone comprehensive static and dynamic security audits to guarantee zero credential exposure before flipping the GitHub repository to public status.

## 2. Incremental Development Milestones

### Increment 1: Core Frontend & UI Design System (Completed Early)

- **Modular Design System:** Hand-crafted DOTON\_DESIGN\_SYSTEM.md using Vanilla CSS variables for high-fidelity dark glassmorphism, gold accents, and multi-theme switching (Night, Day, Sunset).
- **React Component Architecture:** Developed modular components including Header, HeroBanner, FilterBar, AlbumGrid, AlbumDetailModal, and AudioPlayerBar.
- **Persistent Audio Context:** Integrated HTML5 audio player supporting uninterrupted playback while navigating filters and opening detail dialogs, with fallback iTunes preview URL resolution.

### Increment 2: Full-Stack Integration & Database Schema (Completed)

- **Relational Database Architecture:** Modeled and deployed albums, tracks, and recommendations schemas on Supabase PostgreSQL with cascade deletion constraints.
- **Hybrid Data Resilience Layer:** Built dynamic REST fetch logic in supabaseClient.js featuring automatic fallback to local seed data (citypopData.js) during network disruptions.

- **Community Recommendations Pipeline:** Implemented reactive submission modal sending new recommendations directly to the Supabase database with optimistic UI updates.

Increment 3: Features & Pre-Public Security Lockdown (Completed Ahead of Schedule)

- **Featured YouTube Video Header Banner:** Replaced static image placeholder in HeroBanner.jsx with a 500px responsive 16:9 video frame embedding official City Pop video streams (<https://youtu.be/VtRIRJ0tBRc>).
- **Newsletter Subscription Engine:** Built an intuitive newsletter subscription form with real-time email validation, local state persistence, and automatic Supabase PostgreSQL synchronization.
- **Dedicated Album Curation Request Modal:** Developed SuggestAlbumModal.jsx separating the 'Ask What Album To Add Next' flow from community recommendations with automated mailto: dispatch.
- **Authentic Vinyl Cover Resolver & Bug Fixes:** Fixed fuzzy title matching in audioResolver.js and added high-res 600x600 vinyl artwork mappings for *Variety*, *Ride on Time*, *Sea Breeze*, *Timely!!*, and *Pocket Park*.
- **Pre-Public Security Lockdown:** Listed .env in .gitignore, shipped clean .env.example, enforced Row Level Security (RLS) policies on Supabase tables, and completed SECURITY-CHECKLIST.md.

3. Pre-Public Security Lockdown Audit Table

The table below details the security verification items completed prior to making the project repository public:

| Security Check Item                 | Status | Verification & Evidence                                                                                     |
|-------------------------------------|--------|-------------------------------------------------------------------------------------------------------------|
| <code>`.env`</code> Ignored in Git  | Yes    | Exposed in <code>`.gitignore`</code> (lines 14–15); <code>`git status`</code> clean.                        |
| <code>`.env.example`</code> Shipped | Yes    | Shipped with placeholder URLs and anon keys only.                                                           |
| No Hardcoded Secrets in Code        | Yes    | All keys read dynamically via <code>`import.meta.env`</code> ; zero hardcoded keys.                         |
| Clean Git Commit History            | Yes    | Verified via <code>`git log -p`</code> regex search; zero exposed strings.                                  |
| Supabase Row Level Security (RLS)   | Yes    | RLS policies enabled on all tables ( <code>`albums`</code> , <code>`tracks`</code> , <code>`recs`</code> ). |
| Parameterized Database Queries      | Yes    | Queries strictly use Supabase JS builder methods ( <code>`.select()`</code> , <code>`.insert()`</code> ).   |
| GitHub Actions Workflows            | N/A    | No <code>`.github/workflows`</code> present in repository.                                                  |
| SECURITY-CHECKLIST.md Completed     | Yes    | Authored in <code>`project/`</code> directory with full row evidence.                                       |

## 4. Engineering & Resource Distribution

Due to starting the project early, development was structured around high manual code craftsmanship paired with strategic AI assistance: **70% Manual Engineering** (hand-crafted CSS design system, state lifting, audio player context, Supabase SQL schema design, RLS security rules) and **30% AI Assistance** (accelerated boilerplate generation, seed script drafting, and initial styling templates).

## 5. Conclusion & Next Steps

With early commencement and rigorous security lockdown completed, the project repository is fully prepared for public publication and final grading evaluation. Future remaining tasks involve final UI visual checks and hosting deployment.